

Chapter 6

The Laplace Transform

6.1 The Laplace Transform

Objectives

- To understand and be able to compute the **Laplace transform** of a function $f(t)$, which can be accomplished by computing

$$\mathcal{L}(f)(s) = F(s) = \int_0^{\infty} e^{-st} f(t) dt,$$

provided the integral converges.

- To understand that if we know the Laplace transform $F(s)$ of a function, we can recover the original function using the inverse Laplace transform, $\mathcal{L}^{-1}(F(s))(t)$.
- To understand that both the Laplace transform and inverse Laplace transform are linear operators.

Consider the electrical circuit governed by the differential equation

$$LI'' + RI' + \frac{1}{C}I = E'(t).$$

The voltage function, $E'(t)$, might have discontinuities. For example, the voltage in the circuit can be periodically turned on and off. The previous methods that we have used to solve second order linear differential equations may not apply here. However, the **Laplace transform**, an integral transform, gives a method of solving such equations.

As a second example, let us consider a population of fish that is governed by exponential growth,

$$\begin{aligned}\frac{dP}{dt} &= kP \\ P(0) &= P_0,\end{aligned}$$

and suppose that we wish to determine the effects of seasonal fishing. In other words, harvesting will not be continuous. For example, we might only allow

fishing at a constant rate r during the first half of the year,

$$H(t) = \begin{cases} r & 0 \leq t \leq 1/2, \\ 0 & 1/2 < t < 1, \\ r & 1 \leq t \leq 3/2, \\ 0 & 3/2 < t < 2, \\ \vdots & \end{cases}$$

Our initial value problem now becomes

$$\begin{aligned} \frac{dP}{dt} &= kP - H \\ P(0) &= P_0, \end{aligned}$$

It should be clear that we need some additional tools to analyze differential equations possessing discontinuous terms.

Given an initial value problem

$$\begin{aligned} ay'' + by' + cy &= g(t) \\ y(0) &= y_0 \\ y'(0) &= y'_0, \end{aligned}$$

the idea is to use the Laplace transform to change the differential equation into an equation that can be solved algebraically and then transform the algebraic solution back into a solution of the differential equation. Surprisingly, this method will even work when g is a discontinuous function, provided the discontinuities are not too bad.

6.1.1 Definition of the Laplace Transform

We shall define the **Laplace transform** of a function $f(t)$ by

$$\mathcal{L}(f)(s) = F(s) = \int_0^\infty e^{-st} f(t) dt,$$

provided this integral converges. The Laplace transform of a function has many nice properties, especially with respect to the derivative of f . However, before we investigate these properties, let us compute several Laplace transforms.

Example 6.1.1

1. Let $f(t) = k$, where k is a constant. Then

$$\mathcal{L}(f)(s) = \int_0^\infty e^{-st} k dt = \frac{k}{s},$$

for $s > 0$.

2. For $f(t) = t$, the Laplace transform is

$$\mathcal{L}(f)(s) = \int_0^\infty e^{-st} t dt = \frac{1}{s^2},$$

for $s > 0$.

3. If $f(t) = e^{at}$ with $t > 0$, then

$$\mathcal{L}(f)(s) = \int_0^\infty e^{-st} e^{at} dt = \int_0^\infty e^{-(s-a)t} dt = \left[-\frac{e^{-(s-a)t}}{s-a} \right]_0^\infty.$$

Noting that $e^{-(s-a)t} \rightarrow 0$ as $t \rightarrow \infty$ for $s > a$, we find that

$$\mathcal{L}(e^{at}) = \frac{1}{s-a}$$

for $s > a$.

□

Example 6.1.2 The Laplace transform of a function does not always exist, even for functions that are infinitely differentiable. For example, let $f(t) = e^{t^2}$. Then for any real number s , we know that

$$e^{t^2} e^{-st} = e^{t(t-s)} > 1,$$

for $t > s$ with $t \geq 0$; hence,

$$\int_0^b e^{t^2} e^{-st} dt = \int_0^b e^{t(t-s)} dt \geq b.$$

Thus, the integral

$$\int_0^\infty e^{t^2} e^{-st} dt$$

does not converge.

□

The Laplace transform, however, does exist in many cases. In [Subsection 6.1.3](#), we will show that the Laplace transform of a function exists provided the function does not grow too quickly and does not possess bad discontinuities.

6.1.2 Properties of the Laplace Transform

One of the most important properties of the Laplace transform is linearity. That is,

$$\mathcal{L}(\alpha f + \beta g)(s) = \alpha \mathcal{L}(f)(s) + \beta \mathcal{L}(g)(s),$$

where $\alpha, \beta \in \mathbb{R}$ provided the Laplace transforms of f and g exist. The proof of this statement follows directly from the definition of the Laplace transform and the properties of integration,

$$\begin{aligned} \mathcal{L}(\alpha f + \beta g)(s) &= \int_0^\infty e^{-st} [\alpha f(t) + \beta g(t)] dt \\ &= \lim_{b \rightarrow \infty} \int_0^b e^{-st} [\alpha f(t) + \beta g(t)] dt \\ &= \alpha \lim_{b \rightarrow \infty} \int_0^b e^{-st} f(t) dt + \beta \lim_{b \rightarrow \infty} \int_0^b e^{-st} g(t) dt \\ &= \alpha \mathcal{L}(f)(s) + \beta \mathcal{L}(g)(s) \end{aligned}$$

We state the linearity of the Laplace transform as a theorem.

Theorem 6.1.3 *Let f and g be two functions whose Laplace transforms $\mathcal{L}(f)(s)$ and $\mathcal{L}(g)(s)$ exist for $s > a$ and $s > b$, respectively. Then for any real constants α and β and $s > \max\{a, b\}$*

$$\mathcal{L}(\alpha f + \beta g)(s) = \alpha \mathcal{L}(f)(s) + \beta \mathcal{L}(g)(s).$$

Transforms of functions having the linearity property are called **linear operators**.

Example 6.1.4 Recall that $\sinh at = (e^{at} - e^{-at})/2$. Provided $a > 0$, linearity makes it very easy to compute the Laplace transform of $\sinh at$. By [Example 6.1.1](#),

$$\mathcal{L}(\sinh at) = \frac{1}{2}\mathcal{L}(e^{at}) - \frac{1}{2}\mathcal{L}(e^{-at}) = \frac{1}{2}\left(\frac{1}{s-k} - \frac{1}{s+k}\right).$$

Thus, for $s > k > 0$,

$$\mathcal{L}(\sinh at) = \frac{k}{s^2 - k^2}.$$

□

The Laplace transform of discontinuous functions also exist, provided the discontinuities are not too bad. We say that a function f is **piecewise continuous** on an interval $[a, b]$ if f satisfies the following conditions.

1. There are a finite number of discontinuities of f on $[a, b]$.
2. If $c \in [a, b]$ is a discontinuity, then the one-sided limits

$$\lim_{t \rightarrow c^-} f(t) \quad \text{and} \quad \lim_{t \rightarrow c^+} f(t)$$

both exist. We also assume that $\lim_{t \rightarrow a^+} f(t)$ and $\lim_{t \rightarrow b^-} f(t)$ exist.

We say that a function is **piecewise continuous** on an interval $[0, \infty)$, if it is piecewise continuous on the interval $[0, b]$ for all $b > 0$. The types of discontinuities that we are describing are sometimes called **jump discontinuities**. If a function is piecewise continuous, then the discontinuities are not too bad.

Example 6.1.5 One of the simplest piecewise continuous functions is

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0. \end{cases}$$

The function $u(t)$ has a jump discontinuity at $t = 0$. It follows very quickly that $\mathcal{L}(u(t)) = 1/s$ for $s > 0$.

If we define the unit step function at a to be

$$u_a(t) = u(t - a) = \begin{cases} 0, & t < a \\ 1, & t \geq a, \end{cases}$$

then we have a jump discontinuity at $t = a$ ([Figure 6.1.6](#)). If $a > 0$, then

$$\mathcal{L}(u_a(t)) = \int_0^\infty e^{-st} u_a(t) dt = \int_a^\infty e^{-st} dt = \lim_{b \rightarrow \infty} \left[-\frac{e^{-st}}{s} \right]_a^b = \frac{e^{-sa}}{s},$$

where $s > 0$ and $a > 0$. A function of the form $u_a(t) = u(t - a)$ is called a **step function** or **Heaviside function**, named for the British engineer Oliver Heaviside.

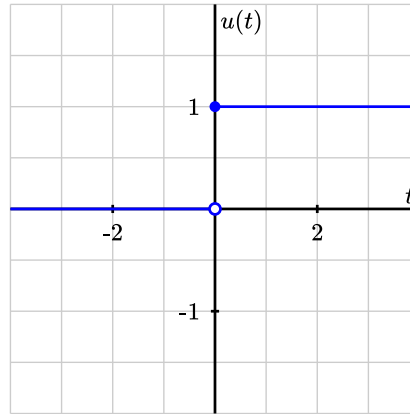


Figure 6.1.6 Unit step function for $a = 0$

□

Example 6.1.7 Given a function $f(t)$, consider the new function

$$g(t) = u_a(t)f(t - a).$$

To obtain the function g from f , we shift the graph of the f to the right by a units and let $g(t) = 0$ for $0 \leq t < a$ (see Figure 6.1.8). Let us compute the Laplace transform of g ,

$$\mathcal{L}(g) = \int_0^\infty g(t)e^{-st} dt = \int_a^\infty f(t - a)e^{-st} dt.$$

If we make the substitution $w = t - a$, then

$$\mathcal{L}(g) = \int_0^\infty f(w)e^{-s(w+a)} dt = e^{-sa} \int_0^\infty f(w)e^{-sw} dt = e^{-sa} \mathcal{L}(f).$$

In other words, if $\mathcal{L}(f) = F(s)$, then $\mathcal{L}(u_a(t)f(t - a)) = e^{-as}F(s)$.

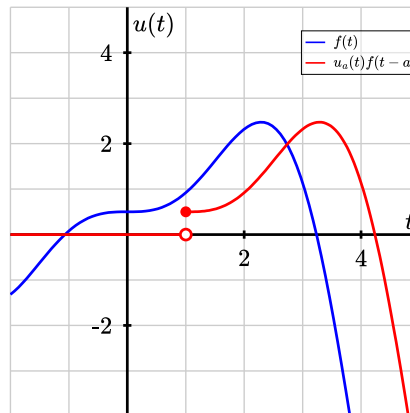


Figure 6.1.8 $u_a(t)f(t - a)$, where $a = 1$

□

Activity 6.1.1 Finding Laplace Transforms. Find the Laplace transform for each of the functions below. Recall that $\cosh t = (e^t + e^{-t})/2$ and $\sinh t = (e^t - e^{-t})/2$

(a) $f(t) = t^6$

(b) $f(t) = t^7 e^{-t}$

(c) $f(t) = 3 \sin^2(\sqrt{2}t)$

(d) $f(t) = e^{3t} \sin 2t$

(e) $f(t) = 2t^2 \cosh t$

6.1.3 Existence and Uniqueness of the Laplace Transform

If we are to use Laplace transforms to study differential equations, we would like to know which functions actually have Laplace transforms. Furthermore, if two functions have the same Laplace transform, we can ask if the functions must be the same. In other words, we wish to know if the Laplace transform of a function exists and is unique. We can answer both of these questions affirmatively if the function $f(t)$ is piecewise continuous on $[0, \infty)$ and does not grow too quickly as $t \rightarrow \infty$. We say a function $f(t)$ is **exponentially bounded** on $[0, \infty)$ if there exist constants $M \geq 0$ and a such that

$$|f(t)| \leq M e^{at},$$

for all t in $[0, \infty)$. In other words, the graph of f must lie between the curves $y = M e^{at}$ and $y = -M e^{at}$. The following two theorems tell us that Laplace transforms exist and are unique for piecewise continuous, exponentially bounded functions on the interval $[0, \infty)$. We leave the proofs of these theorems as exercises.

Theorem 6.1.9 *If f is a piecewise continuous, exponentially bounded function defined $[0, \infty)$, then the Laplace transform of f ,*

$$\mathcal{L}(f)(s) = F(s) = \int_0^\infty e^{-st} f(t) dt$$

exists.

Theorem 6.1.10 *Let f and g be two piecewise continuous exponentially bounded functions with Laplace transforms, $F(s)$ and $G(s)$, respectively. Suppose that $F(s) = G(s)$ for all $s > c$, where c is some positive number. Then $f(t) = g(t)$ for all $t \geq 0$.*

In light of [Theorem 6.1.10](#), it now makes sense to define the **inverse Laplace transform** of a function $F(s)$, which we will denote by $\mathcal{L}^{-1}(F(s))(t)$, as the function $f(t)$ whose unique Laplace transform is $F(s)$. Furthermore, the inverse Laplace transform is linear,

$$\mathcal{L}^{-1}(\alpha F + \beta G) = \alpha \mathcal{L}^{-1}(F) + \beta \mathcal{L}^{-1}(G).$$

6.1.4 Finding Laplace Transforms and Inverse Transforms

To use the method of Laplace transforms effectively, we need either tables or computer software such as Sage so that we can easily find Laplace transforms and inverse Laplace transforms.

Table 6.1.11 Table of Laplace Transforms

$f(t) = \mathcal{L}^{-1}(F(s))$	$F(s) = \mathcal{L}(f(t))$
1	$1/s, s > 0$
e^{at}	$1/(s-a), s > a$
$t^n, n \in \mathbb{N}$	$n!/s^{n+1}, s > 0$
$t^a, a > -1$	$\Gamma(a+1)/s^{a+1}, s > 0$
$\sin at$	$a/(s^2 + a^2), s > 0$
$\cos at$	$s/(s^2 + a^2), s > 0$
$\sinh at$	$a/(s^2 - a^2), s > a $
$\cosh at$	$s/(s^2 - a^2), s > a $
$e^{at} \sin bt$	$b/[(s-a)^2 + b^2], s > a$
$e^{at} \cos bt$	$(s-a)/[(s-a)^2 + b^2], s > a$
$t^n e^{at}, n \in \mathbb{N}$	$n!/(s-a)^{n+1}, s > a$
$u_c(t)$	$e^{-cs}/s, s > 0$
$u_c(t)f(t-c)$	$e^{-cs}F(s)$
$e^{ct}f(t)$	$F(s-c)$
$f(ct)$	$(1/c)F(s/c), c > 0$
$\delta(t-c) = \delta_c(t)$	e^{-cs}
$f'(t)$	$sF(s) - f(0)$
$f''(t)$	$s^2F(s) - sf(0) - f'(0)$

We can also use Sage to find Laplace transforms and inverse transforms (see [Subsection 6.1.8](#)).

Activity 6.1.2 Finding Inverse Transforms. Find the inverse Laplace transform for each of the following functions.

(a) $F(s) = \frac{2}{s^2}$

(b) $F(s) = \frac{1}{s^4}$

(c) $F(s) = \frac{2}{(s-3)^2 + 4}$

(d) $F(s) = \frac{2s^2}{(s^2+1)(s-1)}$

(e) $F(s) = \frac{2s-13}{s(s^2-4s+13)}$

6.1.5 Important Lessons

- Many initial value problems have discontinuous forcing terms.
- The **Laplace transform** of a function $f(t)$ by

$$\mathcal{L}(f)(s) = F(s) = \int_0^\infty e^{-st} f(t) dt,$$

provided the integral converges.

- If f is a piecewise continuous, exponentially bounded function defined $[0, \infty)$, then the Laplace transform of f ,

$$\mathcal{L}(f)(s) = F(s) = \int_0^\infty e^{-st} f(t) dt$$

exists.

- The Laplace transform is a linear operator; i.e.,

$$\mathcal{L}(\alpha f + \beta g)(s) = \alpha \mathcal{L}(f)(s) + \beta \mathcal{L}(g)(s),$$

where $\alpha, \beta \in \mathbb{R}$ provided the Laplace transforms of f and g exist.

- If we know the Laplace transform $F(s)$ of a function, we can recover the original function using the inverse Laplace transform of a function $\mathcal{L}^{-1}(F(s))(t)$.

- The inverse Laplace transform is linear,

$$\mathcal{L}^{-1}(\alpha F + \beta G) = \alpha \mathcal{L}^{-1}(F) + \beta \mathcal{L}^{-1}(G).$$

- If $\mathcal{L}(f) = F(s)$, then $\mathcal{L}(u_c(t)f(t-c)) = e^{-sc}F(s)$.

- If $c \geq 0$, we define the **Heaviside function** to be

$$u_c(t) = \begin{cases} 0 & t < c \\ 1 & t \geq c. \end{cases}$$

The Laplace transform of u_c is

$$\mathcal{L}(u_c(t))(s) = \frac{e^{-cs}}{s}.$$

6.1.6 Reading Questions

1. Does the Laplace transform of a function always exist? If so, why? If not, given an example?
2. What is the Heaviside function? Explain.

6.1.7 Exercises

Finding Laplace transforms. Find the Laplace transform for each of the functions in [Exercise Group 6.1.7.1–8](#). Recall that $\cosh t = (e^t + e^{-t})/2$ and $\sinh t = (e^t - e^{-t})/2$.

- | | |
|------------------------------------|-------------------------|
| 1. $f(t) = \sin at$ | 2. $f(t) = \cos at$ |
| 3. $f(t) = e^{at} \sin bt$ | 4. $f(t) = \cosh at$ |
| 5. $f(t) = 2 \cos at - 3 \sinh at$ | 6. $f(t) = te^{at}$ |
| 7. $f(t) = t \sin at$ | 8. $f(t) = t^2 \cos at$ |

Finding inverse Laplace transforms. Find the inverse Laplace transform for each of the functions in [Exercise Group 6.1.7.9–16](#). You will find partial fraction decomposition very useful.

- | | |
|------------------------------------|---------------------------------------|
| 9. $F(s) = \frac{1}{s(s+1)}$ | 10. $F(s) = s^{-3/2}$ |
| 11. $F(s) = \frac{1}{s+7}$ | 12. $F(s) = \frac{2}{s^2+9}$ |
| 13. $F(s) = \frac{5s-4}{2s^2+s-1}$ | 14. $F(s) = \frac{2s^2-s+4}{s^3+4s}$ |
| 15. $F(s) = \frac{1}{(s+2)^3}$ | 16. $F(s) = \frac{1}{(s+1)^2(s^2-9)}$ |

17. Prove [Theorem 6.1.9](#): If f is a piecewise continuous, exponentially bounded function defined $[0, \infty)$, then the Laplace transform of f ,

$$\mathcal{L}(f)(s) = F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

exists.

18. Let f be a function whose Laplace transform $\mathcal{L}(f)(s) = F(s)$ exists for $s > a$. Prove that

$$\mathcal{L}(e^{ct} f(t))(s) = F(s - c)$$

for $s > a + c$.

19. Prove [Theorem 6.1.10](#): Suppose that f and g are piecewise continuous, exponentially bounded functions defined $[0, \infty)$, with Laplace transforms of $F(s)$ and $G(s)$, respectively. If $F(s) = G(s)$, then $f(t) = g(t)$.
20. If f is a piecewise continuous, exponentially bounded function defined $[0, \infty)$, prove that the Laplace transform of f ,

$$\mathcal{L}(f)(s) = F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

exists.

21. Define the **gamma function** to be

$$\Gamma(x) = \int_0^{\infty} e^{-t} t^{x-1} dt$$

for $x > 0$. The gamma function is very useful for expressing the Laplace transform of the function t^α .

- (a) Show that $\Gamma(1) = 1$.
- (b) Prove that $\Gamma(x + 1) = x\Gamma(x)$ and deduce that $\Gamma(n + 1) = n!$ for $n \in \mathbb{N}$.
- (c) Show that the Laplace transform of $f(t) = t^\alpha$ is

$$\mathcal{L}(f)(s) = \frac{\Gamma(\alpha + 1)}{s^{\alpha+1}}.$$

If n is a positive integer, then $\mathcal{L}(t^n)(s) = n!/s^{n+1}$.

6.1.8 Laplace Transforms with Sage

Computer algebra systems have now replaced tables of Laplace transforms just as the calculator has replaced the slide rule. It is easy to calculate Laplace transforms with Sage. For example, suppose that we wish to compute the Laplace transform of $f(x) = t^3 e^t - \cos t$. We can use the Sage command `laplace`.

```
s = var("s")
t = var("t")
f = t^3*exp(t) - cos(t)
f.laplace(t,s)
```

```
-s/(s^2 + 1) + 6/(s - 1)^4
```

To recover the original function, we can use the Sage command `inverse_laplace`. Suppose that

$$F(s) = -\frac{s}{s^2 + 1} + \frac{6}{(s - 1)^4}.$$

```
s = var("s")
t = var("t")
F = -s/(s^2 + 1) + 6/(s - 1)^4
F.inverse_laplace(s,t)
```

```
t^3*exp(t) - cos(t)
```

We can even use Sage to find the Laplace transform of piecewise-defined functions. Suppose that

$$f(t) = \begin{cases} 1, & 0 \leq t \leq 3 \\ 0, & t > 3. \end{cases}$$

The Sage command to define a piecewise-defined function is `piecewise`.

```
t = var("t")
f1(t) = 1
f2(t) = 0
f = piecewise([[0, 3], f1], [(3, infinity), f2]])
f.plot((t, 0, 6), thickness=3, color="red",
       axes_labels=['$t$', '$f(t)$'])
```

Sage can be used to find the Laplace transform of a piecewise-defined function.

```
s = var("s")
t = var("t")
f1(t) = 1
f2(t) = 0
f = piecewise([[0, 3], f1], [(3, infinity), f2]])
f.laplace(t,s)
```

6.1.9 Exercises

1. Use Sage to find the Laplace transform of

$$f(t) = \begin{cases} -t, & 0 \leq t \leq 1 \\ 2, & t > 1. \end{cases}$$

2. Use Sage to find the Laplace transforms for each of the functions in [Exercise Group 6.1.7.1–8](#).

3. Use Sage to find the inverse Laplace transform for each of the functions in [Exercise Group 6.1.7.9–16](#).

6.2 Solving Initial Value Problems

Objectives

- To understand how the Laplace transform acts on derivatives.
- To understand how the Laplace transform can be used to solve initial value problems such as

$$\begin{aligned} ay''(t) + by'(t) + cy(t) &= f(t) \\ y(0) &= y_0 \\ y'(0) &= y'_0 \end{aligned}$$

even when $f(t)$ is discontinuous.

There is no need for Laplace transforms when solving many initial value problems. If we consider the initial value problem

$$\begin{aligned} y'' + 7y' + 10y &= 0 \\ y(0) &= 1 \\ y'(0) &= 1, \end{aligned}$$

the methods described in [Chapter 4](#) work quite well. The characteristic polynomial of $y'' + 7y' + 10y = 0$ is

$$r^2 + 7r + 10 = (r + 2)(r + 5),$$

and the general solution of the differential equation must be

$$y(t) = c_1 e^{-2t} + c_2 e^{-5t}.$$

Applying the initial conditions, we find that the solution to our initial value problem is

$$y(t) = 2e^{-2t} - e^{-5t}.$$

However, suppose that we have a harmonic oscillator with a discontinuous forcing term,

$$\begin{aligned} y'' + 2y' + 5y &= h(t) \\ y(0) = y'(0) &= 0, \end{aligned}$$

where $h(t)$ is given by

$$h(t) = \begin{cases} 7 & t < 5 \\ 0 & t \geq 5. \end{cases}$$

The previous techniques that we described might prove cumbersome to solve such an initial value problem. We can easily imagine such equations arising in physics or engineering.

Fortunately, Laplace transforms forms behave very nicely with respect to derivatives. We can use Laplace transforms to transform an initial value problem into an algebraic equation. Once the algebraic equation is solved, we can use the inverse transform to obtain the solution to our original initial value problem.

6.2.1 Laplace Transforms of the Derivative

Suppose that we have linear differential equation with constant coefficients

$$ay''(t) + by'(t) + cy(t) = f(t)$$

and initial conditions $y(0) = y_0$ and $y'(0) = y'_0$. We can take the Laplace transform of both sides to obtain

$$a\mathcal{L}(y''(t)) + b\mathcal{L}(y'(t)) + c\mathcal{L}(y(t)) = \mathcal{L}(f(t)).$$

Notice that we have used the fact that the Laplace transform is a linear operator (Theorem 6.1.3). To proceed further, we need to know the Laplace transform of the derivative of a function.

Theorem 6.2.1 *Let $y = y(t)$ be a piecewise continuous, exponentially bounded function and assume that y' is also exponentially bounded. Then for large values of s*

$$\mathcal{L}(y')(s) = s\mathcal{L}(y)(s) - y(0) = sY(s) - y(0),$$

where $Y(s)$ is the Laplace transform of y .

Proof. We can evaluate the Laplace transform of y' by using integration by parts,

$$\begin{aligned} \mathcal{L}(y')(s) &= \int_0^\infty y'(t)e^{-st} dt \\ &= \lim_{b \rightarrow \infty} \left[e^{-st}y(t) \Big|_0^b + s \int_0^b y(t)e^{-st} dt \right] \\ &= \lim_{b \rightarrow \infty} [e^{-sb}y(b) - y(0) + s\mathcal{L}(y)(s)]. \end{aligned}$$

We claim that $\lim_{b \rightarrow \infty} e^{-sb}y(b) = 0$. Since, y is exponentially bounded, there exist constants $M \geq 0$ and a such that $|y(t)| \leq Me^{at}$, for all t in $[0, \infty)$. Thus,

$$|e^{-sb}y(b)| \leq Me^{-(s-a)b}.$$

The right-hand side of this inequality as $b \rightarrow \infty$ for $s > a$. Thus,

$$\mathcal{L}(y')(s) = s\mathcal{L}(y)(s) - y(0).$$

■

The Laplace transform also behave nicely with respect to higher order derivatives.

Theorem 6.2.2 *Let $y = y(t)$ and $y'(t)$ be piecewise continuous, exponentially bounded functions and assume that y'' is exponentially bounded. Then for large values of s*

$$\mathcal{L}(y'')(s) = s^2\mathcal{L}(y)(s) - sy(0) - y'(0) = s^2Y(s) - sy(0) - y'(0),$$

where $Y(s)$ is the Laplace transform of y . In general, if $y = y(t)$ and all of its derivatives up to order $k-1$ are piecewise continuous, exponentially bounded functions and $y^{(k)}$ is piecewise continuous, then

$$\begin{aligned} \mathcal{L}(y^{(k)})(s) &= s^k\mathcal{L}(y)(s) - s^{k-1}y(0) - \dots - sy^{(k-2)}(0) - y^{(k-1)}(0) \\ &= s^kY(s) - s^{k-1}y(0) - s^{k-2}y'(0) - \dots - sy^{(k-2)}(0) - y^{(k-1)}(0). \end{aligned}$$

We can use Theorem 6.2.1 and Theorem 6.2.2 to solve initial value problems.

Example 6.2.3 Consider the initial value problem

$$\begin{aligned}y'' + 7y' + 10y &= 0 \\ y(0) &= 1 \\ y'(0) &= 1.\end{aligned}$$

By [Theorem 6.2.1](#) and [Theorem 6.2.2](#)

$$\begin{aligned}\mathcal{L}(y')(s) &= s\mathcal{L}(y)(s) - y(0) = sY(s) - y(0) \\ \mathcal{L}(y'')(s) &= s^2\mathcal{L}(y)(s) - sy(0) - y'(0) = s^2Y(s) - sy(0) - y'(0).\end{aligned}$$

where $Y(s)$ is the Laplace transform of y . If we take the Laplace transform of $y'' + 7y' + 10y$, we have

$$\begin{aligned}\mathcal{L}(y'' + 7y' + 10y) &= \mathcal{L}(y'') + 7\mathcal{L}(y') + 10\mathcal{L}(y) \\ &= [s^2Y(s) - sy(0) - y'(0)] + 7[sY(s) - y(0)] + 10Y(s) \\ &= [s^2Y(s) - s - 1] + 7[sY(s) - 1] + 10Y(s).\end{aligned}$$

Since $\mathcal{L}(0)$, we have the algebraic equation

$$[s^2Y(s) - s - 1] + 7[sY(s) - 1] + 10Y(s) = 0.$$

Solving for $Y(s)$, we get

$$Y(s) = \frac{s+8}{s^2+7s+10} = \frac{s+8}{(s+5)(s+2)} = \frac{2}{s+2} - \frac{1}{s+5},$$

where we have used partial fractions to get the last expression.

Since the Laplace transform of e^{at} is $1/(s-a)$, we know that

$$\mathcal{L}^{-1}\left(\frac{1}{s-a}\right) = e^{at}.$$

We can now solve our initial value problem,

$$y(t) = 2e^{-2t} - e^{-5t},$$

and our solution agrees with the one that we found at the beginning of this section. $\square$

Activity 6.2.1 Solving Linear Differential Equations with Laplace Transforms. Use Laplace Transforms to solve each of the following initial value problems.

- (a) $x' - 5x = e^{5t}$, $x(0) = 0$
- (b) $x'' - x' - 6x = 0$, $x(0) = 2$, $x'(0) = -1$
- (c) $x'' + 4x = \sin 3t$, $x(0) = 0$, $x'(0) = 0$

6.2.2 Discontinuous Functions

If $c \geq 0$ and we define the function

$$u_c(t) = \begin{cases} 0 & t < c \\ 1 & t \geq c, \end{cases}$$

recall that the Laplace transform of u_c is given by

$$\mathcal{L}(u_c(t))(s) = \int_0^\infty u_c(t)e^{-st} dt = \frac{e^{-cs}}{s}.$$

We can use this information to solve initial value problems with discontinuous functions.

Example 6.2.4 Consider the initial value problem

$$\begin{aligned} y' + y &= u_3(t) \\ y(0) &= 1. \end{aligned}$$

If we take the Laplace transform of both sides of $y' + y = u_3(t)$, we obtain

$$sY(s) - y(0) + Y(s) = \frac{e^{-3s}}{s}.$$

Using the fact that $y(0) = 1$ and solving for $Y(s)$, we get

$$Y(s) = \frac{1}{s+1} + \frac{e^{-3s}}{s(s+1)}.$$

Therefore,

$$y(t) = \mathcal{L}^{-1}\left(\frac{1}{s+1}\right) + \mathcal{L}^{-1}\left(\frac{e^{-3s}}{s(s+1)}\right).$$

The inverse Laplace transform of the first term is

$$\mathcal{L}^{-1}\left(\frac{1}{s+1}\right) = e^{-t}.$$

To compute the inverse Laplace transform of the second term, recall from [Example 6.1.7](#) that if $\mathcal{L}(f) = F(s)$, then

$$\mathcal{L}(u_a(t)f(t-a)) = e^{-as}F(s).$$

Using partial fractions to obtain

$$\frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1}.$$

Hence,

$$\mathcal{L}^{-1}\left(\frac{e^{-3s}}{s(s+1)}\right) = \mathcal{L}^{-1}\left(\frac{e^{-3s}}{s}\right) - \mathcal{L}^{-1}\left(\frac{e^{-3s}}{s+1}\right) = u_3(t) - \mathcal{L}^{-1}\left(\frac{e^{-3s}}{s+1}\right).$$

If $g(t) = u_3(t)e^{-(t-3)}$, then the Laplace transform of $g(t)$ is

$$\mathcal{L}(g) = e^{-3s}\mathcal{L}(e^{-t}) = \frac{e^{-3s}}{s+1}.$$

Thus,

$$y(t) = e^{-t} + u_3(t)\left(1 - e^{-(t-3)}\right).$$

□

6.2.3 Forced Harmonic Oscillators

Example 6.2.5 Consider the forced harmonic oscillator

$$\begin{aligned}y'' + 4y &= 3 \cos t \\ y(0) &= y'(0) = 0.\end{aligned}$$

Taking the Laplace transform of both sides of the equation $y'' + 4y = 3 \cos t$, we obtain

$$\mathcal{L}(y'') + 4\mathcal{L}(y) = 3\mathcal{L}(\cos t)$$

or

$$s^2 Y(s) - sy(0) - y'(0) + 4Y(s) = \frac{3s}{s^2 + 1},$$

where $\mathcal{L}(y)(s) = Y(s)$. Substituting the initial conditions and solving for Y , we have

$$Y(s) = \frac{3s}{(s^2 + 4)(s^2 + 1)} = \frac{-s}{s^2 + 4} + \frac{s}{s^2 + 1},$$

where the last expression was obtained using partial fractions. Taking the inverse Laplace transform, we have our solution

$$y(t) = \mathcal{L}^{-1}\left(\frac{-s}{s^2 + 4}\right) + \mathcal{L}^{-1}\left(\frac{s}{s^2 + 1}\right) = -\cos 2t + \cos t.$$

□

Example 6.2.6 Now let us consider a harmonic oscillator with discontinuous forcing,

$$\begin{aligned}y'' + 2y' + 5y &= h(t) \\ y(0) &= y'(0) = 0,\end{aligned}$$

where $h(t)$ is given by

$$h(t) = \begin{cases} 5 & t < 7 \\ 0 & t \geq 7. \end{cases}$$

That is, $h(t) = 5(1 - u_7(t))$. We may consider this to be a mass-spring system sliding on a table, where the mass is one unit, the spring constant is 5, and the damping coefficient is 2. When $t < 7$ the table is tilted so that gravity provides a force of 5 units when stretching the spring. At time $t = 7$, the table is suddenly returned to the level position.

Taking the Laplace transform of both sides of $y'' + 2y' + 5y = h(t)$, we obtain

$$[s^2 Y(s) - sy(0) - y'(0)] + 2[sY(s) - y(0)] + 5Y(s) = \mathcal{L}(h),$$

where $\mathcal{L}(y)(s) = Y(s)$. Substituting the initial conditions and evaluating the Laplace transform on the right, we have

$$(s^2 + 2s + 5)Y(s) = 5\left(\frac{1}{s} - \frac{e^{-7s}}{s}\right).$$

Solving for $Y(s)$, we have

$$Y(s) = \frac{5}{s(s^2 + 2s + 5)} - \frac{5e^{-7s}}{s(s^2 + 2s + 5)}$$

and

$$y = \mathcal{L}^{-1} \left(\frac{5}{s(s^2 + 2s + 5)} - \frac{5e^{-7s}}{s(s^2 + 2s + 5)} \right).$$

Using partial fractions, we can rewrite the first term as

$$\frac{5}{s(s^2 + 2s + 5)} = \frac{1}{s} - \frac{s + 2}{s^2 + 2s + 5}.$$

The inverse Laplace transform of $1/s$ is 1. To find the inverse Laplace transform of the second term, we complete the square of the denominator,

$$\begin{aligned} \frac{s + 2}{s^2 + 2s + 5} &= \frac{s + 2}{(s + 1)^2 + 4} \\ &= \frac{s + 1}{(s + 1)^2 + 4} + \frac{1}{(s + 1)^2 + 4} \\ &= \frac{s + 1}{(s + 1)^2 + 4} + \frac{1}{2} \frac{2}{(s + 1)^2 + 4}. \end{aligned}$$

Consequently,

$$\mathcal{L}^{-1} \left(\frac{s + 2}{s^2 + 2s + 5} \right) = e^{-t} \cos 2t + \frac{1}{2} e^{-t} \sin 2t = e^{-t} \left(\cos 2t + \frac{1}{2} \sin 2t \right).$$

and

$$\mathcal{L}^{-1} \left(\frac{5}{s(s^2 + 2s + 5)} \right) = \mathcal{L}^{-1} \left(\frac{1}{s} - \frac{s + 2}{s^2 + 2s + 5} \right) = 1 - e^{-t} \left(\cos 2t + \frac{1}{2} \sin 2t \right).$$

We can compute the inverse Laplace transform of

$$\frac{5e^{-7s}}{s(s^2 + 2s + 5)}$$

using the Heaviside function $u_7(t)$ and the inverse Laplace transform that we just calculated to obtain

$$\mathcal{L}^{-1} \left(\frac{5e^{-7s}}{s(s^2 + 2s + 5)} \right) = u_7(t) \left(1 - e^{-(t-7)} \left(\cos 2(t-7) + \frac{1}{2} \sin 2(t-7) \right) \right).$$

Therefore, the solution to our original initial value problem is

$$y(t) = 1 - e^{-t} \left(\cos 2t + \frac{1}{2} \sin 2t \right) - u_7(t) \left(1 - e^{-(t-7)} \left(\cos 2(t-7) + \frac{1}{2} \sin 2(t-7) \right) \right).$$

□

Activity 6.2.2 Solving Differential Equations with Laplace Transforms with Discontinuous Forcing Functions. Use Laplace Transforms to solve each of the following initial value problems.

(a) $x' + x = u_3(t)$, $x(0) = 2$

(b) $x'' + x = h(t)$, $x(0) = 0$, $x'(0) = 0$ where

$$h(t) = \begin{cases} 0 & t < 1 \\ 2 & t \geq 1. \end{cases}$$

6.2.4 Important Lessons

- Using the Laplace transform including how the transform behaves, we can solve initial value problems such as

$$ay''(t) + by'(t) + cy(t) = f(t)$$

$$y(0) = y_0$$

$$y'(0) = y'_0$$

even when $f(t)$ is discontinuous.

- Let $y = y(t)$ be a piecewise continuous, exponentially bounded function and assume that y' is also exponentially bounded. Then for large values of s

$$\mathcal{L}(y')(s) = s\mathcal{L}(y)(s) - y(0) = sY(s) - y(0),$$

where $Y(s)$ is the Laplace transform of y .

- Let $y = y(t)$ and $y'(t)$ be piecewise continuous, exponentially bounded functions and assume that y'' is exponentially bounded. Then for large values of s

$$\mathcal{L}(y'')(s) = s^2\mathcal{L}(y)(s) - sy(0) - y'(0) = s^2Y(s) - sy(0) - y'(0),$$

where $Y(s)$ is the Laplace transform of y . In general, if $y = y(t)$ and all of its derivatives up to order $k-1$ are piecewise continuous, exponentially bounded functions and $y^{(k)}$ is piecewise continuous, then

$$\begin{aligned}\mathcal{L}(y^{(k)})(s) &= s^k\mathcal{L}(y)(s) - s^{k-1}y(0) - \dots - sy^{(k-2)}(0) - y^{(k-1)}(0) \\ &= s^kY(s) - s^{k-1}y(0) - s^{k-2}y'(0) - \dots - sy^{(k-2)}(0) - y^{(k-1)}(0).\end{aligned}$$

6.2.5 Reading Questions

- Describe in your own words how the Laplace transform can be used to solve an initial value problem.
- Explain what an exponentially bounded function is. Give an example.

6.2.6 Exercises

Solving Initial Value Problems. Solve the initial problems in [Exercise Group 6.2.6.1–8](#) using the Laplace transform.

- $y'' - 2y' - 3y = 3e^{2t}$, $y(0) = 1$, $y'(0) = 0$
- $y'' - y' - 2y = 4x^2$, $y(0) = -1$, $y'(0) = 1$
- $\frac{d^2x}{dx^2} - 6\frac{dx}{dt} + 25x = 64e^{-t}$, $x(0) = 1$, $x'(0) = -2$
- $y'' + 16y = 2\sin 2t$, $y(0) = 1$, $y'(0) = 0$
- $y'' + 16y = 2\sin 4t$, $y(0) = 1$, $y'(0) = 0$
- $y'' + 2y' + y = 2e^{-t}$, $y(0) = -1$, $y'(0) = 3$
- $y'' + 6y' + 8y = \cos 3t$, $y(0) = -2$, $y'(0) = 1$
- $u'' + \omega_0^2 u = \cos \omega t$, $\omega^2 \neq \omega_0^2$, $u(0) = 1$, $u'(0) = -1$
- Find the solution of the initial value problem

$$2y'' + y' + 2y = g(t)$$

$$y(0) = y'(0) = 0$$

where $g(t)$ is defined by

$$g(t) = u_5(t) - u_{20}(t) = \begin{cases} 1, & 5 \leq t < 20, \\ 0, & 0 \leq t < 5 \text{ and } t \geq 20. \end{cases}$$

10. Let $y = y(t)$ and $y'(t)$ be piecewise continuous, exponentially bounded functions and assume that y'' is exponentially bounded.

(a) Prove that

$$\mathcal{L}(y'')(s) = s^2 \mathcal{L}(y)(s) - sy(0) - y'(0) = s^2 Y(s) - sy(0) - y'(0)$$

for large values of s , where $Y(s)$ is the Laplace transform of y .

- (b) If $y = y(t)$ and all of its derivatives up to order $k-1$ are piecewise continuous, exponentially bounded functions and $y^{(k)}$ is piecewise continuous, prove that

$$\begin{aligned} \mathcal{L}(y^{(k)})(s) &= s^k \mathcal{L}(y)(s) - s^{k-1}y(0) - \dots - sy^{(k-2)}(0) - y^{(k-1)}(0) \\ &= s^k Y(s) - s^{k-1}y(0) - s^{k-1}y(0) - \dots - sy^{(k-2)}(0) - y^{(k-1)}(0). \end{aligned}$$

11. Prove [Theorem 6.2.2](#).

12. Let $f(t)$ be a piecewise continuous function for $t \geq 0$. In addition, suppose that $f(t)$ satisfies the condition $|f(t)| \leq Me^{ct}$ with $t \geq 0$.

(a) Prove that

$$\mathcal{L}\left(\int_0^t f(\tau) d\tau\right) = \frac{1}{s} \mathcal{L}(f(t)) = \frac{F(s)}{s}.$$

(b) Show that

$$\mathcal{L}^{-1}\left(\frac{F(s)}{s}\right) = \int_0^t f(\tau) d\tau$$

follows from (a).

13. **Solving Linear Systems Using Laplace Transforms.** Consider the linear system

$$\begin{aligned} u' + u - v &= 0 \\ v' - u + v &= 2 \\ u(0) &= 1, \quad v(0) = 2. \end{aligned}$$

If we let $\mathcal{L}(u(s)) = U(s)$ and $\mathcal{L}(v(s)) = V(s)$ and take the Laplace transform of the two differential equations, we obtain

$$\begin{aligned} [sU(s) - 1] + U(s) - V(s) &= 0 \\ [sV(s) - 2] - U(s) + V(s) &= \frac{2}{s}. \end{aligned}$$

(a) Verify that if we solve the system

$$\begin{aligned} [sU(s) - 1] + U(s) - V(s) &= 0 \\ [sV(s) - 2] - U(s) + V(s) &= \frac{2}{s} \end{aligned}$$

for $U(s)$ and $V(s)$, we obtain

$$U(s) = \frac{s+1}{s^2} \quad \text{and} \quad V(s) = \frac{2s+1}{s^2}.$$

(b) Find $u(t)$ and $v(t)$.

(c) Solve the linear system

$$\begin{aligned}u' + 4u - 6v &= 0 \\v' + 3u - 5v &= 0 \\u(0) &= 3, \quad v(0) = 2\end{aligned}$$

using Laplace transforms.

(d) Solve the linear system

$$\begin{aligned}w' - y &= 0 \\w + y' + z &= 1 \\w - y + z' &= 2 \sin t \\w(0) &= 1, \quad y(0) = 1, \quad z(0) = 1\end{aligned}$$

using Laplace transforms.

6.3 Delta Functions and Forcing

Objectives

- To understand **Impulse forcing**, a term used to describe a very quick push or pull on a system, such as the blow of a hammer or the force of an explosion, and that an impulse function can be described by **Dirac delta function**, $\delta(t)$, which has the properties

$$\begin{aligned}\delta(t) &= 0, \quad t \neq 0; \\ \int_{-\infty}^{\infty} \delta(t) dt &= 1.\end{aligned}$$

- To understand that we can use the Dirac delta function to solve initial value problems such as

$$\begin{aligned}\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 26y &= \delta_4(t) \\ y(0) &= 1 \\ y'(0) &= 0,\end{aligned}$$

or

$$\frac{d^2 y}{dt^2} + p \frac{dy}{dt} + qy = g(t),$$

where $g(t)$ is a function that is very large in a very short time interval.

6.3.1 Impulse Forcing

Impulse forcing is the term used to describe a very quick push or pull on a system, such as the blow of a hammer or the force of an explosion. For example, consider the equation for a damped harmonic oscillator

$$\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 26y = g(t),$$

where $g(t)$ is a function that is very large in a very short time interval, say $|t - t_0| < \tau$ and zero otherwise. The integral

$$I(\tau) = \int_{t_0-\tau}^{t_0+\tau} g(t) dt$$

or since $g(t)$ is zero outside of the interval $|t - t_0| < \tau$

$$I(\tau) = \int_{-\infty}^{\infty} g(t) dt$$

measures the strength or **impulse** of the forcing function $g(t)$. In particular, assume that $t_0 = 0$ and

$$g(t) = d_\tau(t) = \begin{cases} 1/2\tau, & -\tau < t < \tau \\ 0, & \text{otherwise.} \end{cases}$$

It is easy to see that $I(\tau) = 1$ in this case.

Examining this forcing function over shorter and shorter time intervals with τ getting closer and closer to zero, we find that $I(\tau) = 1$ in all cases. Thus,

$$\lim_{\tau \rightarrow 0} d_\tau(t) = 0$$

for $t \neq 0$; however,

$$\lim_{\tau \rightarrow 0} I(\tau) = 1.$$

We can use this information to define the **unit impulse function**, $\delta(t)$, to be the “function” that imparts an impulse of magnitude one at $t = 0$, but is zero for all values of t other than zero. In other words, $\delta(t)$ has the properties

$$\begin{aligned} \delta(t) &= 0, & t &\neq 0; \\ \int_{-\infty}^{\infty} \delta(t) dt &= 1. \end{aligned}$$

Of course, we study no such function in calculus. The “function” δ is an example of what is known as a **generalized function**. We call δ , the **Dirac delta function**.

We can define a unit impulse at a point t_0 by considering the function $\delta(t - t_0)$. In this case,

$$\begin{aligned} \delta(t - t_0) &= 0, & t &\neq t_0; \\ \int_{-\infty}^{\infty} \delta(t - t_0) dt &= 1. \end{aligned}$$

6.3.2 The Laplace Transform of the Dirac Delta Function

Even though the Dirac delta function is not a piecewise continuous, exponentially bounded function, we can define its Laplace transform as the limit of the Laplace transform of $d_\tau(t)$ as $\tau \rightarrow 0$. More specifically, assume that $t_0 > 0$ and

$$\mathcal{L}(\delta(t - t_0)) = \lim_{\tau \rightarrow 0} \mathcal{L}(d_\tau(t - t_0)).$$

Assuming that $t_0 - \tau > 0$, the Laplace transform of $d_\tau(t - t_0)$ is

$$\mathcal{L}(d_\tau(t - t_0)) = \int_0^{\infty} e^{-st} d_\tau(t - t_0) dt$$

$$\begin{aligned}
&= \int_{t_0-\tau}^{t_0+\tau} e^{-st} d_\tau(t-t_0) dt \\
&= \frac{1}{2\tau} \int_{t_0-\tau}^{t_0+\tau} e^{-st} dt \\
&= \frac{1}{2s\tau} e^{-st} \Big|_{t=t_0-\tau}^{t=t_0+\tau} \\
&= \frac{1}{2s\tau} e^{-st_0} (e^{s\tau} - e^{-s\tau}) \\
&= \frac{\sinh s\tau}{s\tau} e^{-st_0}.
\end{aligned}$$

We can use l'Hospital's rule to evaluate $(\sinh s\tau)/s\tau$ as $\tau \rightarrow 0$,

$$\lim_{\tau \rightarrow 0} \frac{\sinh s\tau}{s\tau} = \lim_{\tau \rightarrow 0} \frac{s \cosh s\tau}{s} = 1.$$

Thus,

$$\mathcal{L}(\delta(t-t_0)) = e^{-st_0}.$$

We can extend this result to allow $t_0 = 0$, by

$$\lim_{t_0 \rightarrow 0} \mathcal{L}(\delta(t-t_0)) = \lim_{t_0 \rightarrow 0} e^{-st_0} = 1.$$

Example 6.3.1 Let us now solve the initial value problem

$$\begin{aligned}
\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 26y &= \delta_4(t) \\
y(0) &= 1 \\
y'(0) &= 0.
\end{aligned}$$

We can think of this as a damped harmonic oscillator that is struck by a hammer at time $t = 4$. Let $Y(s) = \mathcal{L}(y)(s)$ and take the Laplace transform of both sides of the differential equation to obtain

$$s^2 Y(s) - sy(0) - y'(0) + 2(sY(s) - y(0)) + 26Y(s) = \mathcal{L}(\delta_4)(s)$$

or

$$s^2 Y(s) - s + 2sY(s) - 2 + 26Y(s) = e^{-4s}.$$

Solving for $Y(s)$, we have

$$Y(s) = \frac{s+2}{s^2+2s+26} + \frac{e^{-4s}}{s^2+2s+26}.$$

The inverse Laplace transform of $Y(s)$ is

$$\begin{aligned}
y &= \mathcal{L} \left(\frac{s+2}{s^2+2s+26} \right) + \mathcal{L} \left(\frac{e^{-4s}}{s^2+2s+26} \right) \\
&= \mathcal{L} \left(\frac{s+2}{(s+1)^2+25} \right) + \frac{1}{5} \mathcal{L} \left(\frac{5e^{-4s}}{(s+1)^2+25} \right) \\
&= e^{-t} \cos 5t + \frac{1}{5} e^{-t} \sin 5t + \frac{1}{5} u_4(t) e^{-(t-4)} \sin(5(t-4)).
\end{aligned}$$

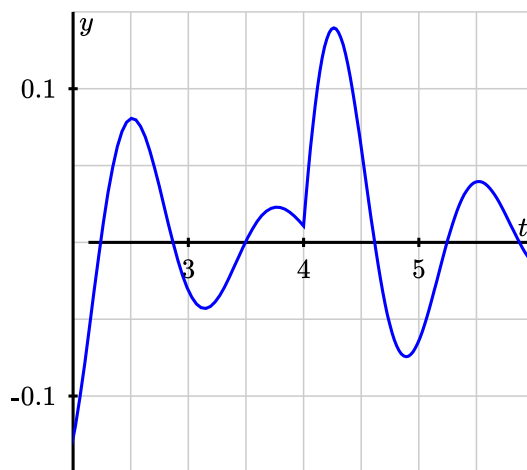


Figure 6.3.2 Solution to $y'' + 2y' + 26y - \delta_4(t) = 0$

□

It is important to notice that we are using the Dirac delta function like an ordinary function. This requires some rigorous mathematics to justify that we can actually do this.

6.3.3 Important Lessons

- **Impulse forcing** is the term used to describe a very quick push or pull on a system, such as the blow of a hammer or the force of an explosion. For example, consider the equation for a damped harmonic oscillator

$$\frac{d^2y}{dt^2} + p\frac{dy}{dt} + qy = g(t),$$

where $g(t)$ is a function that is very large in a very short time interval, say $|t - t_0| < \tau$ and zero otherwise. The integral

$$I(\tau) = \int_{t_0-\tau}^{t_0+\tau} g(t) dt$$

or since $g(t)$ is zero outside of the interval $|t - t_0| < \tau$

$$I(\tau) = \int_{-\infty}^{\infty} g(t) dt$$

measures the strength or **impulse** of the forcing function $g(t)$.

- We define the **unit impulse function**, $\delta(t)$, to be the “function” that imparts an impulse of magnitude one at $t = 0$, but is zero for all values of t other than zero. In other words, $\delta(t)$ has the properties

$$\begin{aligned} \delta(t) &= 0, & t &\neq 0; \\ \int_{-\infty}^{\infty} \delta(t) dt &= 1. \end{aligned}$$

The “function” δ is an example of what is known as a **generalized function**. We call δ , the **Dirac delta function**.

- Similarly, we can define a unit impulse at a point t_0 by considering the function $\delta(t - t_0)$. In this case,

$$\delta(t - t_0) = 0, \quad t \neq 0;$$

$$\int_{-\infty}^{\infty} \delta(t - t_0) dt = 1.$$

- The Laplace transform of the Dirac delta function is

$$\mathcal{L}(\delta(t - t_0)) = e^{-st_0}.$$

We can extend this result to allow $t_0 = 0$, by

$$\lim_{t_0 \rightarrow 0} \mathcal{L}(\delta(t - t_0)) = \lim_{t_0 \rightarrow 0} e^{-st_0} = 1.$$

- We can use the Dirac delta function to solve initial value problems such as

$$\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 26y = \delta_4(t)$$

$$y(0) = 1$$

$$y'(0) = 0,$$

or

$$\frac{d^2 y}{dt^2} + p \frac{dy}{dt} + qy = g(t),$$

where $g(t)$ is a function that is very large in a very short time interval.

6.3.4 Reading Questions

1. What is impulse forcing? Give an example of a physical system where impulse forcing might be useful.
2. Is the Dirac delta function an actual function? Why or why not.

6.3.5 Exercises

Solving Initial Value Problems. Solve the initial problems in [Exercise Group 6.3.5.1–6](#) using the Laplace transform, $\delta(t)$ is the unit impulse function.

1. $2y'' + y' + 2y = \delta(t)$, $y(0) = 0$, $y'(0) = 0$
2. $y'' - y' - 2y = \delta(t - 5)$, $y(0) = 0$, $y'(0) = 0$
3. $\frac{d^2 x}{dx^2} - 6 \frac{dx}{dt} + 25x = \delta(t)$, $x(0) = 1$, $x'(0) = -2$
4. $y'' + 16y = \delta(t)$, $y(0) = 1$, $y'(0) = 0$
5. $y'' + 16y = \delta(t - 4)$, $y(0) = 1$, $y'(0) = 0$
6. $y'' + 2y' + y = \delta(t + 4)$, $y(0) = -1$, $y'(0) = 3$

6.4 Convolution

Objectives

- To understand that if f and g are two piecewise continuous exponentially bounded functions, then we can define the **convolution product** of f and g to be

$$(f * g)(t) = \int_0^t f(t - \tau)g(\tau) d\tau = \int_0^t f(\tau)g(t - \tau) d\tau.$$

- To understand that the convolution product has many properties similar to those of ordinary multiplication.
- To understand that if f and g be two piecewise continuous exponentially bounded functions and $\mathcal{L}(f)(s) = F(s)$ and $\mathcal{L}(g)(s) = G(s)$ for $s \geq a > 0$, then

$$F(s)G(s) = \mathcal{L}(f * g)(s)$$

for $s > a$.

- To understand that it is possible to write a solution for the initial value problem

$$\begin{aligned} ay'' + by' + cy &= g(t) \\ y(0) &= y_0 \\ y'(0) &= y_1. \end{aligned}$$

using the **Convolution Theorem**.

When solving an initial value problem using Laplace transforms, we employed the strategy of converting the differential equation to an algebraic equation. Once the algebraic equation is solved, we can recover the solution to the initial value problem using the inverse Laplace transform. While our strategy is straight forward, finding an inverse Laplace transform can be a hindrance. Often finding the inverse transform involves decomposing a complicated product into the sum for partial fractions. Fortunately, there is a product rule for inverse Laplace transforms. This product rule will allow us to quickly compute solutions of a harmonic oscillator with different forcing functions.

6.4.1 Convolution

If f and g are two piecewise continuous exponentially bounded functions, then we define the **convolution product** of f and g to be

$$(f * g)(t) = \int_0^t f(t - \tau)g(\tau) d\tau = \int_0^t f(\tau)g(t - \tau) d\tau.$$

The convolution product has many properties similar to those of ordinary multiplication.

- Commutivity: $f * g = g * f$.
- Distribution: $f * (g + h) = f * g + f * h$.
- Associativity: $f * (g * h) = (f * g) * h$.
- $0 * f = f * 0 = 0$.

All of these properties can be proven using the definition of convolution and the properties of integration.

There is, however, no multiplicative identity. In other words, $f * 1 \neq f$. For example, suppose that $f(t) = \cos t$. Then

$$(f * 1)(t) = \int_0^t \cos(t - \tau) \cdot 1 \, d\tau = \sin t.$$

Also, it may not be the case that $f * f$ is a nonnegative function.

A key property of the Laplace transform is how convolution products behave.

Theorem 6.4.1 Convolution Theorem. *Let f and g be two piecewise continuous exponentially bounded functions, and suppose that $\mathcal{L}(f)(s) = F(s)$ and $\mathcal{L}(g)(s) = G(s)$ for $s \geq a > 0$. Then*

$$F(s)G(s) = \mathcal{L}(f * g)(s)$$

for $s > a$.

Proof. If

$$\begin{aligned} F(s) &= \int_0^\infty e^{-s\xi} f(\xi) \, d\xi \\ G(s) &= \int_0^\infty e^{-s\tau} g(\tau) \, d\tau, \end{aligned}$$

then

$$\begin{aligned} F(s)G(s) &= \int_0^\infty e^{-s\xi} f(\xi) \, d\xi \int_0^\infty e^{-s\tau} g(\tau) \, d\tau \\ &= \int_0^\infty g(\tau) \left(\int_\tau^\infty e^{-st} f(t - \tau) \, dt \right) d\tau, \end{aligned}$$

where $\xi = t - \tau$ is the change of variable. Reversing the order of integration, we have

$$\int_0^\infty g(\tau) \left(\int_\tau^\infty e^{-st} f(t - \tau) \, dt \right) d\tau = \int_0^\infty e^{-st} \left(\int_0^t f(t - \tau) g(\tau) \, d\tau \right) dt.$$

However, this last expression is just the Laplace transform of $f * g$. ■

6.4.2 Applying the Convolution Theorem

The Convolution Theorem ([Theorem 6.4.1](#)) can be very useful for finding the inverse transforms of products, since

$$F(s)G(s) = \mathcal{L}(f * g)(s).$$

For example, suppose that we wish to find the inverse transform of

$$\frac{1}{s^2(s+1)} = \frac{1}{s^2} \cdot \frac{1}{s+1}.$$

Since $\mathcal{L}[t] = 1/s^2$ and $\mathcal{L}[e^{-t}] = 1/(s+1)$, we know that

$$\mathcal{L}^{-1} \left[\frac{1}{s^2(s+1)} \right] = \mathcal{L}^{-1} \left[\frac{1}{s^2} \right] * \mathcal{L}^{-1} \left[\frac{1}{s+1} \right]$$

$$\begin{aligned}
&= \int_0^t (t-u)e^{-u} du \\
&= t + e^{-t} - 1.
\end{aligned}$$

Of course, we could have used partial fractions to decompose $1/(s^2(s+1))$.

Example 6.4.2 Instead of using partial fractions, let us use the Convolution Theorem ([Theorem 6.4.1](#)) to calculate the inverse Laplace transform of

$$H(s) = \frac{a}{s^2(s^2 + a^2)} = \frac{1}{s^2} \cdot \frac{a}{s^2 + a^2}.$$

The inverse Laplace transform of $1/s^2$ is t , and the inverse Laplace transform of $a/(s^2 + a^2)$ is $\sin at$. Using the Convolution Theorem, the inverse Laplace transform of $H(s)$ is

$$h(t) = \int_0^t (t-\tau) \sin a\tau \, d\tau = \frac{at - \sin at}{a^2}.$$

□

We can also use the Convolution Theorem to solve initial value problems.

Example 6.4.3 Consider the initial value problem

$$\begin{aligned}
y'' + 4y &= g(t) \\
y(0) &= 3 \\
y'(0) &= -1.
\end{aligned}$$

Taking the Laplace transform of both sides of the differential equation and applying the initial conditions, we obtain

$$s^2 Y(s) - 3s + 1 + 4Y(s) = G(s),$$

where $G(s)$ is the Laplace transform of $g(t)$. Solving for $Y(s)$, we have

$$Y(s) = \frac{3s-1}{s^2+4} + \frac{G(s)}{s^2+4} = 3 \frac{s}{s^2+4} - \frac{1}{2} \frac{2}{s^2+4} + \frac{1}{2} \frac{2}{s^2+4} G(s).$$

The last term corresponds to the forcing term of our differential equation. Taking the inverse Laplace transform of both sides and applying the Convolution Theorem, we get

$$y = 3 \cos 2t - \frac{2}{2} \sin 2t + \frac{1}{2} \int_0^t \sin 2(t-\tau) g(\tau) \, d\tau.$$

□

Notice that we did not specify a particular forcing function in [Example 6.4.3](#). In fact, it is possible to write a solution for the initial value problem

$$\begin{aligned}
ay'' + by' + cy &= g(t) \\
y(0) &= y_0 \\
y'(0) &= y_1
\end{aligned}$$

using the Convolution Theorem without knowing what the actual forcing function $g(t)$ is. Taking the Laplace transform of both sides of the differential equation and using the initial conditions, we have

$$(as^2 + bs + c)Y(s) - (as + b)y_0 - ay_1 = G(s)$$

or

$$Y(s) = \Phi(s) + \Psi(s),$$

where

$$\begin{aligned}\Phi(s) &= \frac{(as+b)y_0 + ay_1}{as^2 + bs + c} \\ \Psi(s) &= \frac{G(s)}{as^2 + bs + c}.\end{aligned}$$

Therefore,

$$y = \phi(t) + \psi(t),$$

where $\phi(t) = \mathcal{L}^{-1}(\Phi(s))$ and $\psi(t) = \mathcal{L}^{-1}(\Psi(s))$. Observe that $\phi(t)$ is the solution to the initial value problem

$$\begin{aligned}ay'' + by' + cy &= 0 \\ y(0) &= y_0 \\ y'(0) &= y_1.\end{aligned}$$

Once we have values for a , b , and c , the function $\phi(t)$ is easy to find. To find $\psi(t)$, we first write

$$\Psi(s) = \frac{1}{as^2 + bs + c}G(s) = H(s)G(s),$$

where $H(s) = 1/(as^2 + bs + c)$. If $h(t)$ is the inverse Laplace transform of $H(s)$, then

$$\psi(t) = \mathcal{L}^{-1}(H(s)G(s)) = \int_0^t h(t-\tau)g(\tau) d\tau.$$

If we consider the case where $G(s) = 1$, then $g(t) = \delta(t)$ and $\psi(s) = H(s)$. This means that $h(t)$ is a solution to the initial value problem

$$\begin{aligned}ay'' + by' + cy &= \delta(t) \\ y(0) &= y_0 \\ y'(0) &= y_1.\end{aligned}$$

For this reason, $h(t)$ is sometimes called the **impulse response** of the system.

6.4.3 Important Lessons

- If f and g are two piecewise continuous exponentially bounded functions, then we define the **convolution product** of f and g to be

$$(f * g)(t) = \int_0^t f(t-\tau)g(\tau) d\tau = \int_0^t f(\tau)g(t-\tau) d\tau.$$

- The convolution product has many properties similar to those of ordinary multiplication.
 - Commutivity: $f * g = g * f$
 - Distribution: $f * (g + h) = f * g + f * h$
 - Associativity: $f * (g * h) = (f * g) * h$
 - $0 * f = f * 0 = 0$

There is, however, no multiplicative identity. In other words, $f * 1 \neq f$. Also, it may not be the case that $f * f$ is a nonnegative function.

- Let f and g be two piecewise continuous exponentially bounded functions, and suppose that $\mathcal{L}(f)(s) = F(s)$ and $\mathcal{L}(g)(s) = G(s)$ for $s \geq a > 0$. Then

$$F(s)G(s) = \mathcal{L}(f * g)(s)$$

for $s > a$.

- It is possible to write a solution for the initial value problem

$$\begin{aligned} ay'' + by' + cy &= g(t) \\ y(0) &= y_0 \\ y'(0) &= y_1. \end{aligned}$$

using the Convolution Theorem.

6.4.4 Reading Questions

1. Explain how the convolution product of two functions is useful when solving an initial value problem.

6.4.5 Exercises

Convolution of Two Functions. Calculate the convolution product of $f(t)$ and $g(t)$ in [Exercise Group 6.4.5.1–6](#) using the definition of the convolution of two functions.

1. $f(t) = 2$ and $g(t) = t$
2. $f(t) = t$ and $g(t) = t$
3. $f(t) = t$ and $g(t) = e^{3t}$
4. $f(t) = t$ and $g(t) = \sin t$
5. $f(t) = e^{2t}$ and $g(t) = e^{3t}$
6. $f(t) = e^{3t}$ and $g(t) = e^{2t}$

Convolution and the Inverse Transform. Calculate the inverse Laplace transform of $F(s)$ in [Exercise Group 6.4.5.7–12](#) using the convolution product.

7. $F(s) = \frac{1}{(s-1)(s-3)}$
8. $F(s) = \frac{3}{s(s-1)}$
9. $F(s) = \frac{2}{s^2(s^2+2)}$
10. $F(s) = \frac{1}{s^2+4s-12}$
11. $F(s) = \frac{1}{(s-1)(s-3)}$
12. $F(s) = \frac{1}{(s-1)(s-3)}$

Solving Initial Value Problems. Solve the initial problems in [Exercise Group 6.4.5.13–18](#) using the method described in [Subsection 6.4.2](#).

13. $2y'' + y' + 2y = g(t)$, $y(0) = 0$, $y'(0) = 0$
14. $y'' - y' - 2y = g(t)$, $y(0) = 0$, $y'(0) = 0$
15. $\frac{d^2x}{dx^2} - 6\frac{dx}{dt} + 25x = g(t)$, $x(0) = 1$, $x'(0) = -2$
16. $y'' + 16y = g(t)$, $y(0) = 1$, $y'(0) = 0$
17. $y'' + 16y = g(t)$, $y(0) = 1$, $y'(0) = 0$
18. $y'' + 2y' + y = g(t)$, $y(0) = -1$, $y'(0) = 3$
19. Prove that the distributive property holds for the convolution product. That is, show that $f * g = g * f$.
20. Prove that the convolution product of two functions f and g distributes multiplication over addition. That is, show that $f * (g + h) = f * g + f * h$ for functions f , g , and h .
21. Prove that the convolution product is associative. In other words, show that $f * (g * h) = (f * g) * h$ for functions f , g , and h .
22. Prove that $0 * f = f * 0 = 0$.

6.5 Projects for Laplace Transforms

Project 6.5.1 Project—Tank Interrupt Mixing. Water containing 2 kg of salt per liter (L) is poured into a tank at a rate of 1 L/min, and the well-stirred mixture leaves at the same rate. After 10 min, the process is stopped and fresh water is poured into the tank at a rate of 1 L/min, with the new mixture leaving at 1 L/min. We seek to model this situation in order to determine the amount (kg) of salt in the tank at the end of 20 minutes if there were 100 L of pure water initially in the tank.¹

- (a) Build a mathematical model for the amount of salt in kg in the tank at time t in min with two parts, first over the time interval $[0, 10]$ minutes and then over the time interval $[10, 20]$ minutes. You can assume instantaneous mixing of the two different solutions.
- (b) Build a mathematical model for the amount of salt in kg in the tank at time t in min using the Heaviside or Unit Step functions to represent the sudden change in input water types and apply Laplace Transforms and Inverse Laplace Transforms to determine the amount of salt in the tank over the entire interval $[0, 20]$ minutes. Then compare your results with Activity 1.

¹This project is adapted from Norman Loney (2018), “7-040-S-TankInterruptMixing,” <https://www.simiode.org/resources/5103>.